

Wnt/ β -catenin signalling regulates cardiomyogenesis via GATA transcription factors

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Abstract

A functioning heart muscle is required continuously throughout life. During embryonic development the heart muscle tissue differentiates from mesoderm that has heart-forming potential. Heart-forming potential in the embryonic mesoderm is regulated by pro-cardiogenic transcription factors, such as members of the GATA and NK-2 transcription factor families. Subsequent heart muscle differentiation involves the expression of cytoskeletal proteins, including myosins and troponins. Different Wnt signalling pathways have various functions in heart development. So-called 'canonical' (Wnt/ β -catenin-mediated) signalling has a conserved role in vertebrate heart development, regulating and restricting heart development and subsequent heart muscle differentiation. Here we investigated the way in which Wnt/ β -catenin signalling functionally interacts with the GATA family of pro-cardiogenic transcription factors to regulate subsequent heart muscle differentiation. We used whole *Xenopus* embryos as an accessible experimental model system for vertebrate heart development. Our experiments confirmed that activation of Wnt signalling results in reduced *gata* gene expression, as well as reduced gene expression of other pro-cardiogenic transcription factors and heart muscle differentiation markers. Remarkably, we discovered that when GATA function is experimentally restored, the expression of other pro-cardiogenic transcription factors and heart muscle differentiation markers is rescued. These findings, obtained from whole-embryo experiments, show that Wnt signalling regulates heart development at the level of GATA factors, confirming earlier results from tissue-culture experiments. Furthermore, our rescue experiments in *Xenopus* embryos revealed differences in functional activity between the various GATA transcription factors involved in heart development. We discovered that GATA4 is more efficient at reinstating the gene expression of other pro-cardiogenic transcription factors, whereas GATA6 is more potent at promoting the expression of genes associated with terminal heart muscle differentiation. In conclusion, our findings show that the inhibition of heart development by Wnt/ β -catenin signalling during organogenesis is mediated by the loss of expression of GATA pro-cardiogenic transcription factors and reveal functional differences between those GATA factors in heart development.

Key words cardiogenesis; cardiomyocytes; GATA; heart; muscle; Nkx; Wnt signalling.

Introduction

The heart is the first functional organ to develop in vertebrate embryos and heart muscle function is required continuously throughout life. Both the initial induction in the early embryo of cardiogenic mesoderm with heart-forming potential and subsequent heart muscle (cardiomyocyte) differentiation are fundamental developmental steps that are conserved among vertebrate embryos

(reviewed by Cohen et al. 2008). *Xenopus* is a powerful experimental model system for studying these conserved steps in vertebrate heart development (Warkman & Krieg, 2007).

The initial induction of heart development is associated with the expression of two families of pro-cardiogenic transcription factors: NK-2 (e.g. Nkx2.3 and Nkx2.5, Sparrow et al. 2000) and GATA (GATA4, GATA5 and GATA6, reviewed by Patient & McGhee, 2002; Jiang & Evans, 1996). Both types of transcription factor function within a molecular network that regulates cardiogenic tissue identity and the subsequent expression of structural genes needed for cardiomyocyte differentiation, including myosin light chain 2 (MLC2) (Latinkic et al. 2004) and Troponin Ic (TnIc) (Warkman & Atkinson, 2004).

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We recently discovered that Wnt6/ β -catenin signalling functions to restrict heart muscle development during stages of organogenesis (Lavery et al. 2008). Reinstatement of GATA function in tissue-culture experiments relieves such Wnt6/ β -catenin-mediated inhibition of cardiomyocyte development (Afouda et al. 2008). Here we use whole-embryo experiments to confirm that reinstated GATA function overcomes the inhibition of heart development by Wnt/ β -catenin signalling. We also discover that different GATA factors preferentially rescue either pro-cardiogenic transcription factor gene expression or the expression of cardiomyocyte differentiation markers.

Materials and methods

Xenopus embryo culture

Xenopus embryos were harvested as previously described (Hoppler, 2008), injected at the four-cell stage with mRNA (see below), treated with dexamethasone and/or 6-bromo-iridium-3-oxime (BIO) (usually at stage 20; see below) and left to develop to stage 32 when they were fixed in MEMFA [MOPS (0.1 M), EGTA (2 mM), MgSO₄ (1 mM), Formaldehyde 3.7–4%] for whole-mount RNA *in-situ* analysis (see below) or else subjected to RNA extraction using an RNeasy extraction kit (Qiagen) for subsequent quantitative reverse transcription-polymerase chain reaction (qPCR) analysis (see below).

Expression constructs and mRNA injections

β -cateninGR is an inducible activator of nuclear Wnt/ β -catenin signalling consisting of a fusion between the activated β -catenin (pt β -catenin, Yost et al. 1996) and the hormone-binding domain of human glucocorticoid receptor (GR) in the pCS2+ vector (Afouda et al. 2008). β -cateninGR-encoding mRNA was prepared, using an mMessage mMachine High Yield RNA transcription kit (Ambion), from an Asp718-linearized plasmid template with SP6 polymerase. GATA4GR and GATA6GR are inducible constructs of the pro-cardiogenic transcription factors xGATA4 and xGATA6, respectively (Afouda et al. 2005). Embryos were injected at the four-cell stage with the indicated amounts of mRNA into the prospective ventral marginal zone (for the axis duplication assays shown in Fig. 1) or into the prospective cardiogenic mesoderm in the dorsal marginal zone (for studying heart development as in all subsequent figures).

Treatments with 6-bromo-iridium-3-oxime and dexamethasone

To activate the inducible β -cateninGR, GATA4GR and GATA6GR proteins, embryos previously injected with the relevant mRNA and controls were treated with the artificial steroid hormone dexamethasone (2 μ M) at stage 20 of development, unless indicated otherwise. The specific Glycogen Synthase Kinase 3 inhibitor BIO (Meijer et al. 2003; Sato et al. 2004) was used at 6 μ M, unless indicated otherwise, to activate Wnt/ β -catenin signalling (Lavery & Hoppler, 2008b) in embryos at stage 20.

RNA *in-situ* hybridization

Spatial expression of the pro-cardiogenic transcription factor Nkx2.5 and the heart muscle differentiation markers MLC2 and Tnlc was analysed by whole-mount RNA *in-situ* hybridization, performed as previously described (Lavery & Hoppler, 2008a). Digoxigenin-labelled RNA probes were prepared with T7 polymerase, using High Yield Megascript (Ambion) from HindIII-linearized plasmid template for xNkx2.5 (Tonissen et al. 1994), BamHI-linearized plasmid template for xMLC2 (Evans et al. 1995) and NotI-linearized plasmid template for Tnlc (Drysdale et al. 1994).

Quantitative reverse transcription-polymerase chain reaction

The qPCR was used as previously described (Lavery & Hoppler, 2008a) to analyse the expression of genes encoding the pro-cardiogenic transcription factors Nkx2.3, GATA4 and GATA6 (as in Lavery et al. 2008) and of the GATA target Wnt11b (as in Afouda et al. 2008), using the validated polymerase chain reaction primer pairs listed in Table 1.

Results and Discussion

Wnt/ β -catenin signalling can inhibit heart development in early vertebrate embryos. Inhibition of endogenous Wnt signalling by extracellular Wnt inhibitors is required in early embryogenesis to promote cardiogenesis (i.e. *dkk1*, *crescent*; Marvin et al. 2001; Schneider & Mercola, 2001). We recently discovered that endogenous Wnt6 functions to restrict heart muscle development (cardiomyogenesis), during later stages of embryogenesis (i.e. organogenesis), and that this function is mediated by activation of the canonical Wnt/ β -catenin signalling pathway (Lavery et al. 2008).

In order to determine more precisely when, during organogenesis, nuclear Wnt/ β -catenin signalling is capable of inhibiting heart muscle development (see below), we used an artificial β -catenin construct encoding a version of β -catenin activated by dexamethasone (β -cateninGR, see Materials and methods). We used axis duplication assays (reviewed by Hoppler, 2008) to verify the efficacy of β -cateninGR in activating Wnt/ β -catenin signalling and to establish the correct experimental conditions for subsequent experiments (in particular, β -cateninGR mRNA injection levels). This assay revealed that dexamethasone-activated β -cateninGR is capable of activating Wnt/ β -catenin signalling (Fig. 1H,J,L) if sufficient β -cateninGR mRNA is injected (at least 20 pg, Fig. 1H, better results at 30 pg, Fig. 1J,M). The injection of amounts larger than 30 pg has no additional benefit as it does not lead to further increased activity (Fig. 1K,M).

We previously established the efficacy of β -cateninGR in tissue-culture experiments (Afouda et al. 2008). The axis duplication assay shown in Fig. 1 confirmed that β -cateninGR can be successfully used in whole-embryo experiments if injected at the appropriate dose (i.e. maintaining

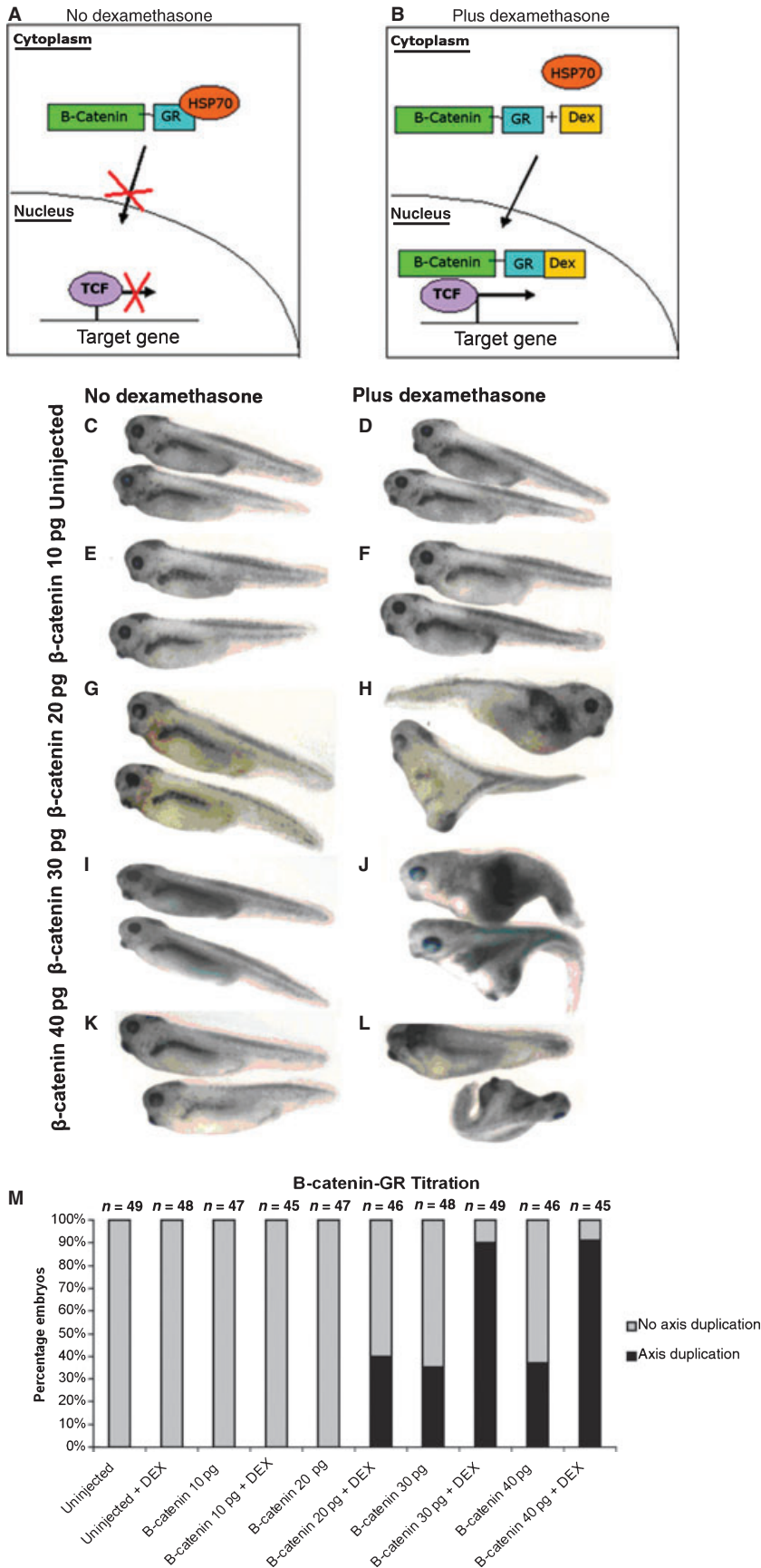


Fig. 1 β -cateninGR is an inducible activator of Wnt signalling. (A,B) Schematic illustration explaining the experimental activation of nuclear Wnt signalling by β -cateninGR and dexamethasone. Without dexamethasone (A), β -cateninGR is retained in the cytoplasm by association with molecular chaperones, such as Heat Shock Protein 70. After dexamethasone treatment (B), β -cateninGR enters the nucleus to regulate Wnt target gene expression in association with T-cell Factor DNA-binding factors. For the axis duplication assay, embryos were either left uninjected (C,D) or injected at the four-cell stage into the prospective ventral side (ventral marginal zone) with 10 pg (E,F), 20 pg (G,H), 30 pg (I,J) or 40 pg (K,L) of mRNA encoding β -cateninGR per blastomere. In order to activate the β -cateninGR construct, uninjected control (D) and mRNA-injected (F,H,J,L) embryos were treated with dexamethasone and compared with control embryos with no dexamethasone (C,E,G,I,K) at tadpole embryo stage 40. Note the development of secondary heads in embryos shown in H,J,L. (M) Percent bar chart summarizing the numerical analysis of the axis duplication assay showing good activation when 30 pg of β -cateninGR mRNA per blastomere is injected. n, number of embryos assayed for each experimental and control manipulation.

Table 1 *Xenopus* polymerase chain reaction primers.

Primer name	Primer sequence 5'–3'	Annealing temperature (°C)	Reference
xODC-F	GTCAATGATGGAGTGTATGGATC	60	http://www.hhmi.ucla.edu/derobertis/
xODC-R	TCCATTCCGCTCTCCTGAGCAC		
xGATA6A-F3	CTGCCACACATCAACAACAAC	58	Lavery et al. (2008)
xGATA6A-R3	GTCATGGAAATTTGGTGACTG		
xGATA4-F	GTGCCACCTATGCAAGCCC	60	Jiang & Evans (1996)
xGATA4-R	TAGACCCACCCGCGGAGAC		
xNkx2.5-F	GAGCTACAGTTGGGTGTGTGGT	62	Sasai et al. (1995)
xNkx2.5-R	GTGAAGCGACTAGGTATGTGTCA		
xNkx-2.3-F3	GTGACAGCCAGTCCTTACACC	60	Lavery et al. (2008)
xNkx-2.3-R3	GACATGAAGGAAGTGGAGTCC		
xMLC2-F	GAGGCATTGAGCTGTATCGA	55	Small et al. (2005)
xMLC2-R	GGACTCCAGAACATGTCTATT		
xTpnlc-F	CCTTGACAGAACTGTCTGAGC	58	Ariizumi et al. (2003)
xTpnlc-R	CAGATTAAGTGCCTTGGAAACG		
xWnt11-F	GAAGTCAAGCAAGTCTGCTGG	60	Ariizumi et al. (2003)
xWnt11-R	GCAGTAGTCAGGGGAACAAACAG		

the requirement for dexamethasone induction). Based on the results of this assay, we decided to inject 30 pg of β -cateninGR mRNA per blastomere (i.e. 60 pg per embryo) in subsequent experiments, always together with controls lacking dexamethasone treatment.

Wnt/ β -catenin signalling restricts cardiogenesis during organogenesis stages

We recently discovered that endogenous Wnt6 restricts heart muscle development during organogenesis stages, probably via canonical Wnt/ β -catenin signalling, as global activation of the pathway throughout the embryo at these stages mimics the Wnt6-mediated inhibition of cardiomyogenesis (Lavery et al. 2008). Targeted injection of our inducible β -cateninGR mRNA provides the opportunity to: (i) confirm the inhibition of cardiogenesis by Wnt/ β -catenin signalling specifically in tissues with heart-forming potential and (ii) determine more precisely at which stage during organogenesis Wnt/ β -catenin signalling can regulate cardiomyogenesis.

We injected β -cateninGR-encoding mRNA into prospective cardiogenic tissue, activated with dexamethasone either before or during the organogenesis stages, and analysed the gene expression of a pro-cardiogenic transcription factor (GATA6, Jiang & Evans, 1996; Peterkin et al. 2003) and a structural gene indicative of heart muscle differentiation (cardiomyogenesis, MLC2; Chambers et al. 1994) (Fig. 2). Irrespective of the developmental stage, the activation of β -cateninGR caused a reduction in the expression of both GATA6 and MLC2. However, the expression of both genes was most reduced when Wnt/ β -catenin signalling was activated at the beginning of organogenesis (stage 20).

Control embryos that were not treated with dexamethasone showed expression levels similar to uninjected embryos.

Our experiments showed that Wnt/ β -catenin signalling is capable of restricting heart muscle development when activated in a relatively tissue-specific manner in the prospective cardiac mesoderm. Although Wnt/ β -catenin signalling can clearly inhibit cardiogenesis at earlier developmental stages (e.g. Marvin et al. 2001; Afouda et al. 2008; Lavery et al. 2008), Fig. 2 shows that Wnt/ β -catenin signalling can still inhibit cardiogenesis at a remarkably late stage in organogenesis, consistent with the role of Wnt6 during heart organogenesis (Lavery et al. 2008). The inducibility of β -cateninGR allowed us to determine more precisely that heart muscle development was particularly susceptible to regulation by Wnt/ β -catenin signalling during the beginning of organogenesis. The results shown in Fig. 2, with Wnt/ β -catenin signalling activated in a relatively tissue-specific manner, are remarkably similar to previous results of less targeted experiments (Lavery et al. 2008) where the Wnt/ β -catenin pathway was induced throughout the embryo by treating entire embryos with a pharmacological activator (Sato et al. 2004).

Still more revealing is the fact that the results were essentially the same for both classes of marker genes analysed (upstream pro-cardiogenic transcription factor GATA6 and terminal cardiomyocyte differentiation marker MLC2). As MLC2 expression requires GATA6 function (Peterkin et al. 2003), our results are consistent with the idea that the regulation of cardiomyocyte differentiation by Wnt signalling is mediated by GATA6 or other pro-cardiogenic transcription factors.

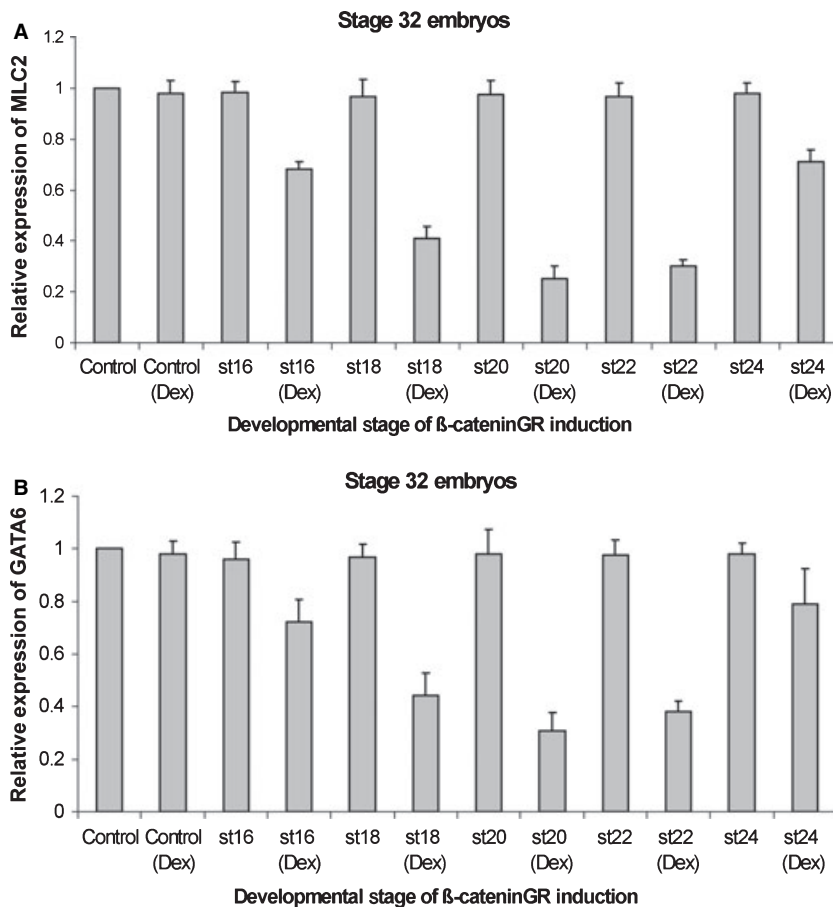


Fig. 2 β -catenin-mediated inhibition of cardiomyogenesis is strongest during the beginning of organogenesis. Analysis of gene expression by qPCR at the end of organogenesis (stage 32) of the cardiogenic transcription factor *gata6* (A) and the heart muscle differentiation marker MLC2 (B). Embryos were injected at the four-cell stage into the two prospective dorsal blastomeres (to target the cardiogenic tissue) with mRNA encoding the inducible β -cateninGR (30 pg per blastomere) to activate Wnt/ β -catenin signalling in a stage-specific manner. Embryos were left untreated (as a control) or treated with dexamethasone to activate β -cateninGR starting at different embryonic stages from mid-neurula (stage 16) to early organogenesis (stage 20) and mid-organogenesis (stage 24) as indicated. Gene expression in embryos with activated β -cateninGR is expressed relative to untreated but β -cateninGR mRNA-injected controls. Statistical analysis was then carried out using a parametric paired sample *t*-test. This analysis reveals that the ability of activated Wnt/ β -catenin signalling to restrict heart development is strongest at the stages at the beginning of organogenesis (stage 20).

Restored GATA function rescues cardiogenesis

Our finding that Wnt/ β -catenin signalling regulates GATA gene expression (Fig. 2) is consistent with earlier findings (Marvin et al. 2001; Afouda et al. 2008; Lavery et al. 2008). GATA4–6 genes encode important transcription factors that are required at the top of the gene regulatory network controlling vertebrate heart development (reviewed by Brewer & Pizzey, 2006). These observations suggest the hypothesis that Wnt/ β -catenin signalling regulates cardiomyogenesis by repressing GATA gene expression. We therefore designed experiments to test this hypothesis in whole embryos. We tested whether reinstating GATA function would rescue aspects of cardiomyogenesis, in a situation where endogenous *gata* gene expression is inhibited by Wnt/ β -catenin signalling. We therefore co-injected β -cateninGR-encoding mRNA and inducible GATA4GR- or GATA6GR-encoding mRNA into prospective cardiogenic tissue (Afouda et al. 2005). In this experimental approach, when we activated β -cateninGR with dexamethasone (and thereby inhibited endogenous *gata* gene expression), we concomitantly reinstated GATA function by activating GATA4GR or GATA6GR function.

Within the molecular network regulating heart development, GATA factors are known to affect the other pro-

cardiogenic transcription factor, Nkx2.5 (e.g. Brewer et al. 2005). We therefore first tested whether activating Wnt/ β -catenin signalling by β -cateninGR resulted in reduced Nkx2.5 expression. The activation of β -cateninGR at early organogenesis (stage 20) resulted in much reduced Nkx2.5 expression (Figs 3G and 4G). However, if we concomitantly reinstated either GATA4 or GATA6 function, strong Nkx2.5 expression was restored (Figs 3H and 4H, respectively).

Nkx2.5 is an important pro-cardiogenic transcription factor (Fu et al. 1998; Grow & Krieg, 1998) and therefore the restoration of Nkx2.5 expression provides good evidence that cardiogenesis is rescued in these embryos. To test more directly whether heart muscle differentiation had been rescued, we analysed whether the expression of the structural genes MLC2 and Tnlc had been affected in a similar way. Indeed, we found that, when GATA4 or GATA6 function was restored in embryos (with concomitant restoration of Nkx2.5 expression as shown above), MLC2 and Tnlc expression was also rescued (Figs 3P,X and 4P,X).

Our results suggest that Wnt/ β -catenin signalling regulates heart development at the level of the *gata* transcription factor genes, as we found that their expression was reduced by activated β -cateninGR (Fig. 2, see also Fig. 5) and cardiomyogenesis was rescued by reinstating their

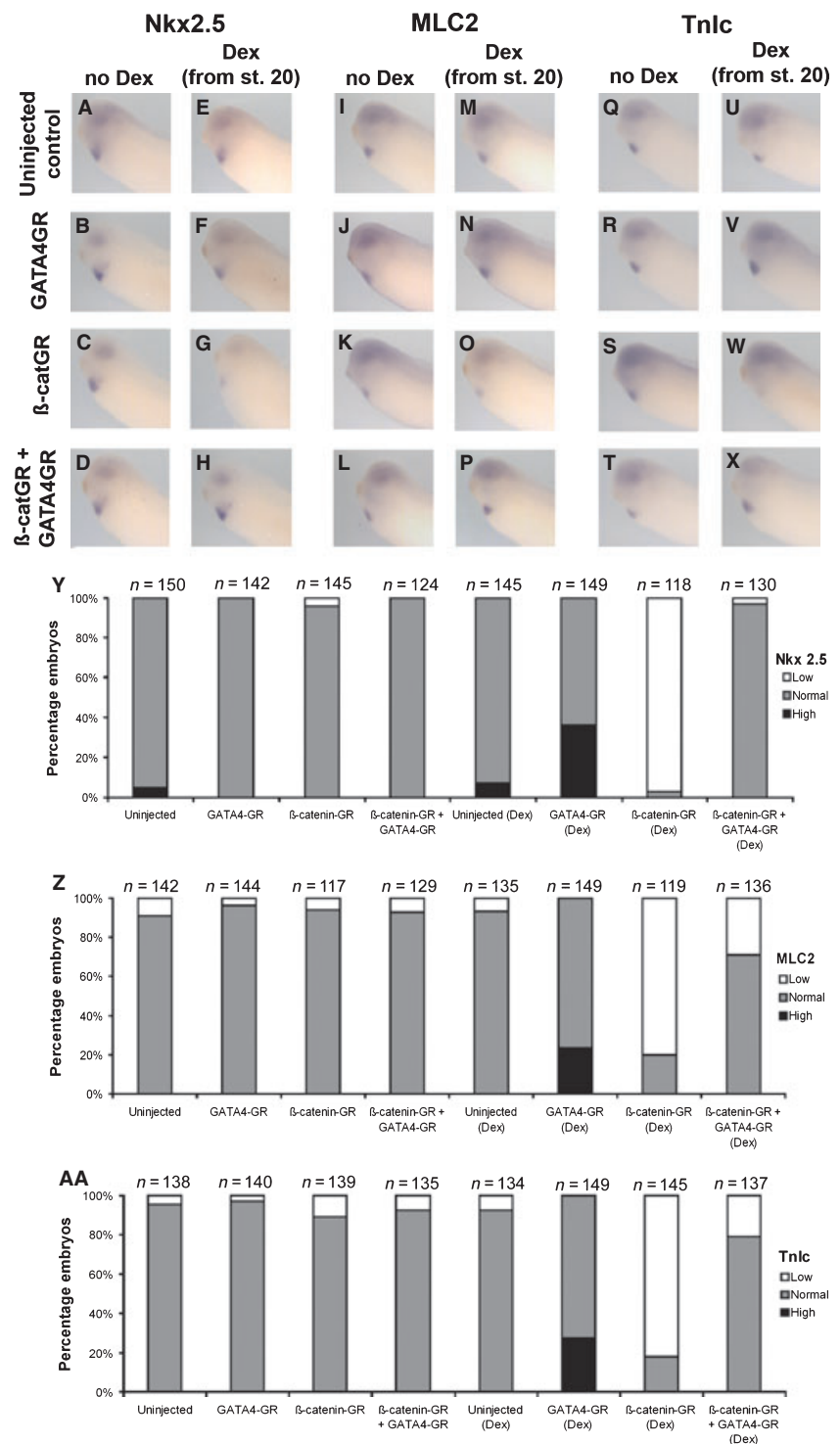


Fig. 3 Stage-specific activation of GATA4 during organogenesis stages rescues the expression of cardiogenic genes after Wnt inhibition. Analysis of Nkx2.5 (A–H), MLC2 (I–P) and Tnnc (Q–X) expression by whole-mount RNA *in-situ* hybridization at stage 32 in uninjected control (A,E,I,M,Q,U), GATA4GR mRNA-injected (100 pg per blastomere) (B,F,J,N,R,V) and β -cateninGR mRNA-injected (C,G,K,O,S,W) embryos and in GATA4GR and β -cateninGR mRNA co-injected embryos (D,H,L,P,T,X), which were left untreated as a control (A–D,I–L,Q–T) or treated with dexamethasone at stage 20 (E–H,M–P,U–X). Note in β -catenin-activated embryos the reduced expression of Nkx2.5 (G), MLC2 (O) and Tnnc (W). When GATA4 function is restored, note the rescued expression of Nkx2.5 (H), MLC2 (P) and Tnnc (X). Bar charts showing the percentage of embryos with high, normal or low Nkx2.5 (Y), MLC2 (Z) and Tnnc (AA) expression in the experiments illustrated in A–X. n, number of embryos assayed for each experimental and control manipulation.

function (Figs 3 and 4). This interpretation of the results of our whole-embryo experiments is consistent with previous findings in tissue-culture experiments (Afouda et al. 2008). However, cardiogenic transcription factors are known to regulate each other and our results confirm earlier findings that GATA transcription factors induce Nkx2.5 expression (e.g. Brewer et al. 2005). There is also evidence that Nkx2.5

in turn regulates *gata* gene expression (Molkentin et al. 2000). Because of this self-enforced regulatory loop involving pro-cardiogenic transcription factors, we cannot currently rule out that Wnt/ β -catenin signalling initially inhibits Nkx2.5 (or another pro-cardiogenic transcription factor), which then indirectly leads to reduced *gata* expression and that the same self-enforced regulatory loop

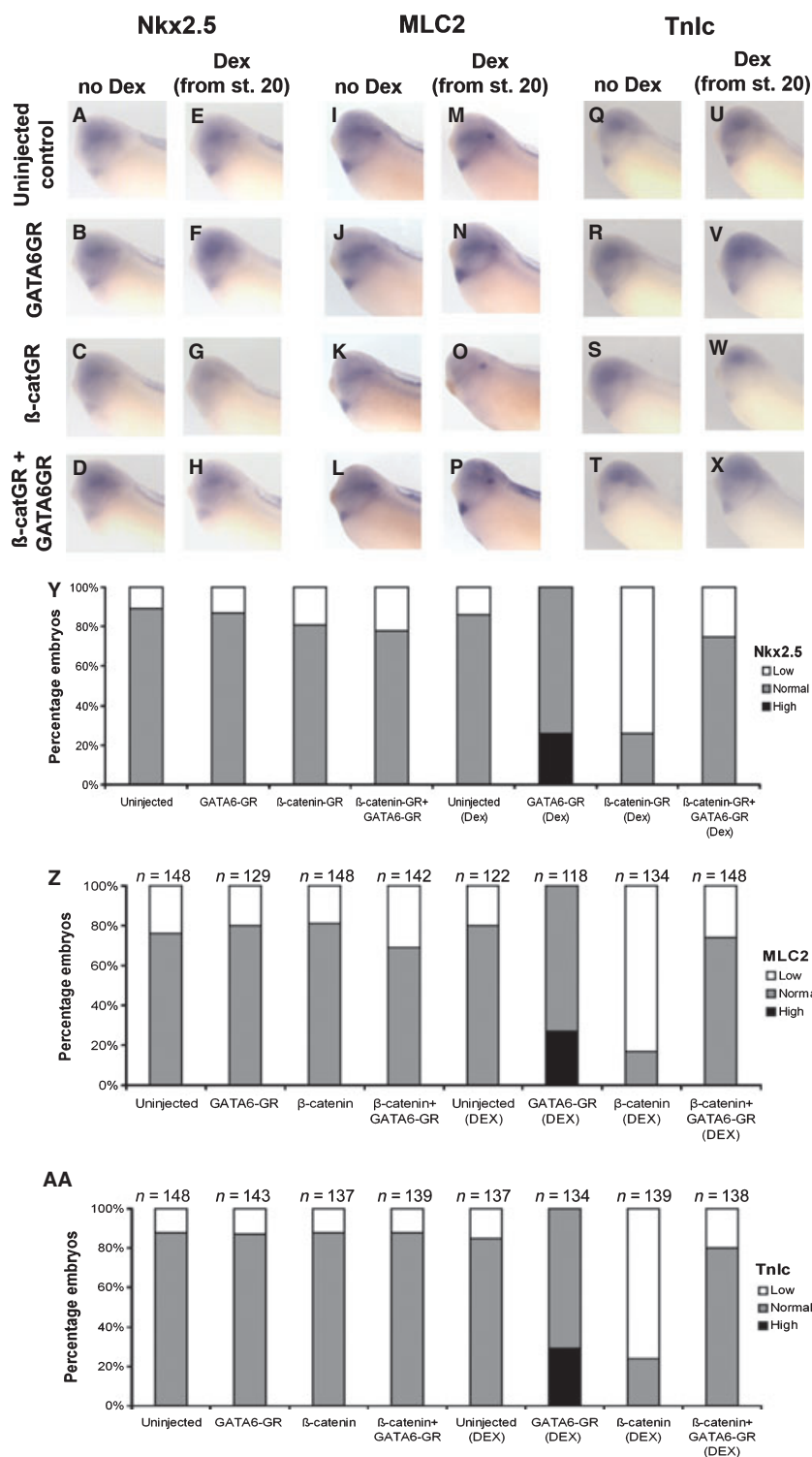


Fig. 4 Stage-specific activation of GATA6 during organogenesis stages rescues the expression of cardiogenic genes after Wnt inhibition. Analysis of Nkx2.5 (A–H), MLC2 (I–P) and Tnlc (Q–X) expression by whole-mount RNA *in-situ* hybridization at stage 32 in uninjected control (A,E,I,M,Q,U), GATA6GR mRNA-injected (100 pg per blastomere) (B,F,J,N,R,V) and β-cateninGR mRNA-injected (C,G,K,O,S,W) embryos and in GATA4GR and β-cateninGR mRNA co-injected embryos (D,H,L,P,T,X), which were left untreated as a control (A–D,I–L,Q–T) or treated with dexamethasone at stage 20 (E–H,M–P,U–X). Note in β-catenin-activated embryos the reduced expression of Nkx2.5 (G), MLC2 (O) and Tnlc (W). When GATA6 function is restored, note the rescued expression of Nkx2.5 (H), MLC2 (P) and Tnlc (X). Bar charts showing percentage of embryos with high, normal or low Nkx2.5 (Y), MLC2 (Z) and Tnlc (AA) expression in the experiments illustrated in A–X. n, number of embryos assayed for each experimental and control manipulation.

ensures that when we reinstate GATA function in our experiments we also reinstate Nkx2.5 expression (and any other cardiogenic transcription factor within this gene regulatory network, which might conceivably be directly regulated by Wnt/β-catenin signalling). Although a much more

detailed molecular analysis of the regulation of expression of these pro-cardiogenic transcription factor genes by Wnt/β-catenin signalling will be required to identify the initial target of Wnt/β-catenin regulation in cardiogenesis, the *gata4–6* genes remain prime candidates. Our results do

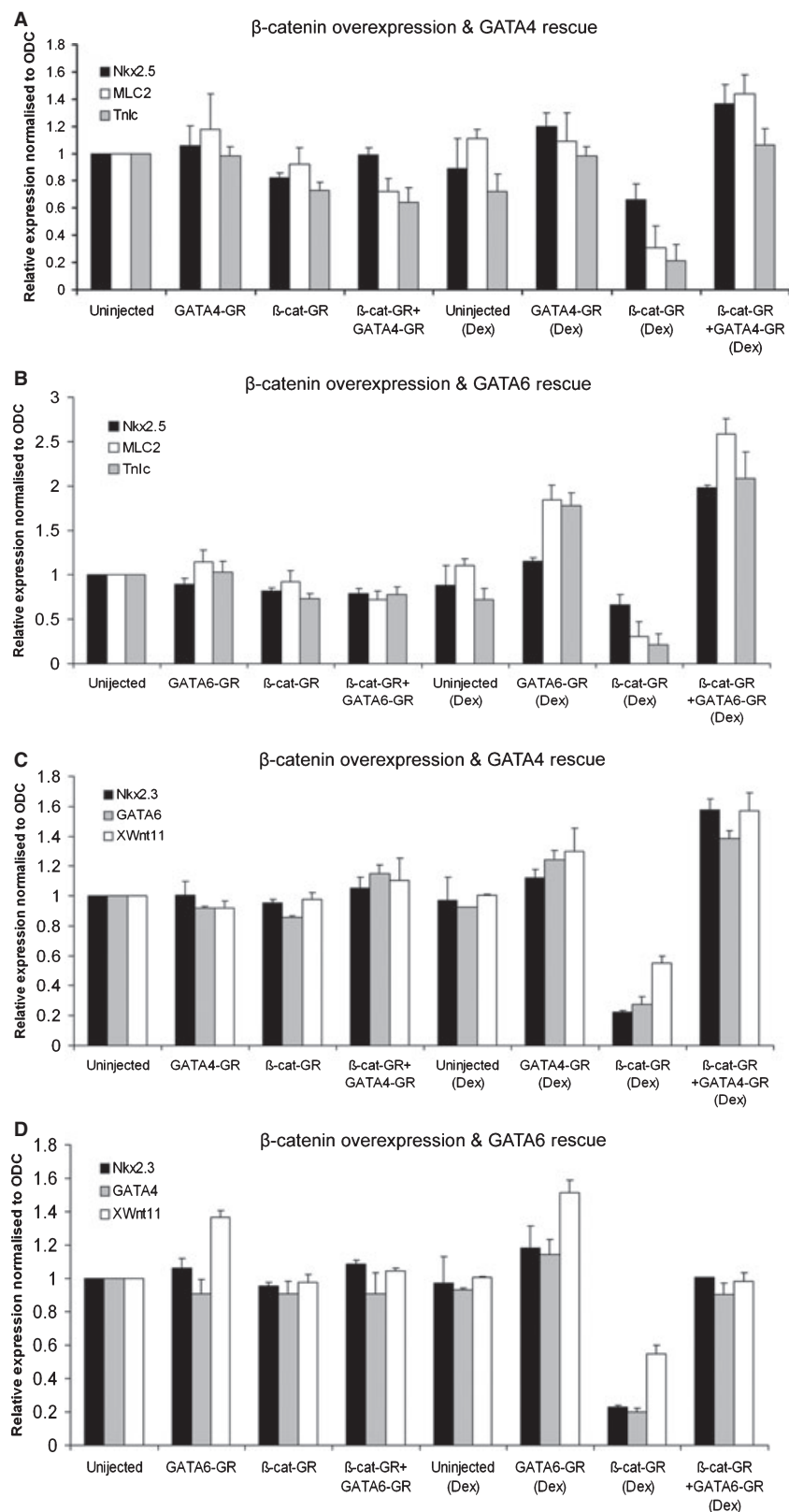


Fig. 5 Wnt/ β -catenin-mediated inhibition of cardiogenic marker gene expression is rescued by GATA4/6 function. Analysis of gene expression for Nkx2.5, MLC2 and Tnnc (A,B) and Nkx2.3, *gata4/6* and Wnt11b (C,D) using qPCR in whole embryos at stage 32. Embryos were injected at the four-cell stage in the dorsal marginal zone of both blastomeres with β -cateninGR mRNA (30 pg per blastomere) alone or as indicated together with GATA4GR (A,C) or GATA6GR mRNA (B,D) (50 pg per blastomere). Embryos were cultured from stage 20 where indicated with dexamethasone to activate injected mRNA constructs. This experiment was repeated twice more before all data were collated. The relative expression of the described cardiac markers was normalized to the housekeeping gene ornithine decarboxylase (ODC) and measured in comparison to control unmanipulated embryos normalized to equal 1 relative expression unit. Statistical analysis was then carried out using a parametric paired sample *t*-test.

show in any case that Wnt/ β -catenin signalling regulates heart development at the level of the pro-cardiogenic transcription factor network.

We wanted to analyse further molecular markers to confirm the results of our experiments and to explore the idea of the self-enforcing regulatory loop for pro-cardiogenic

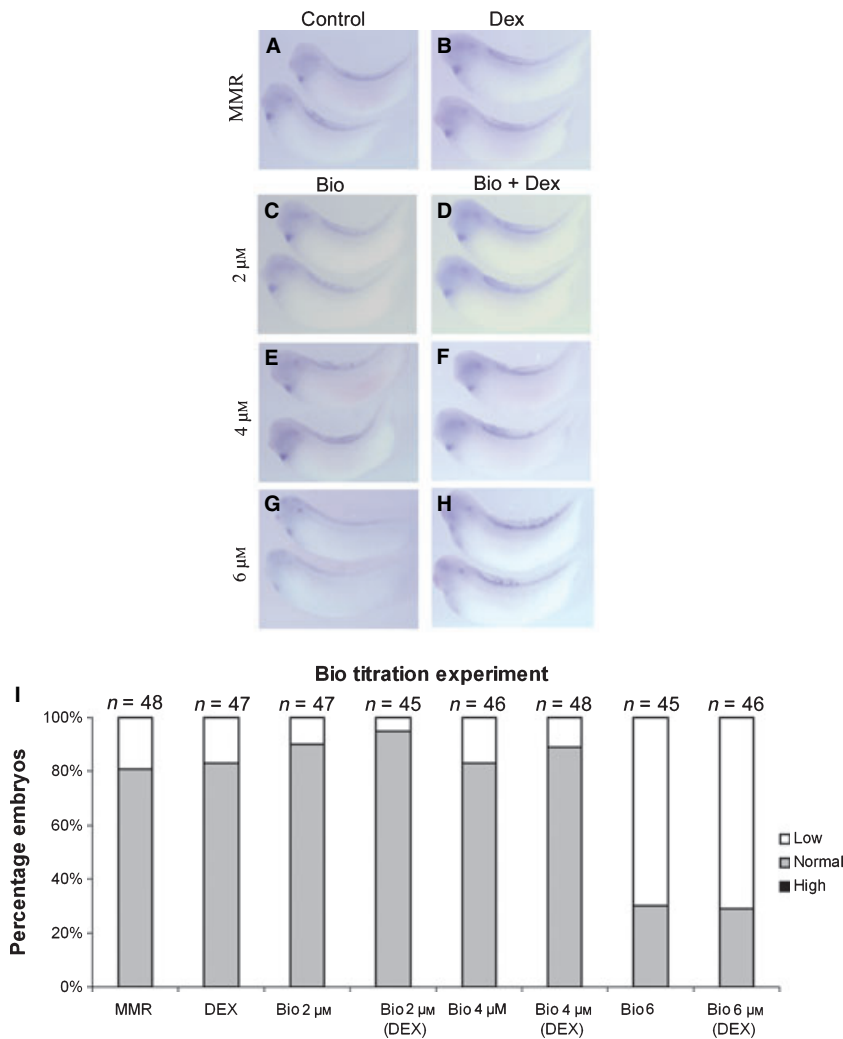
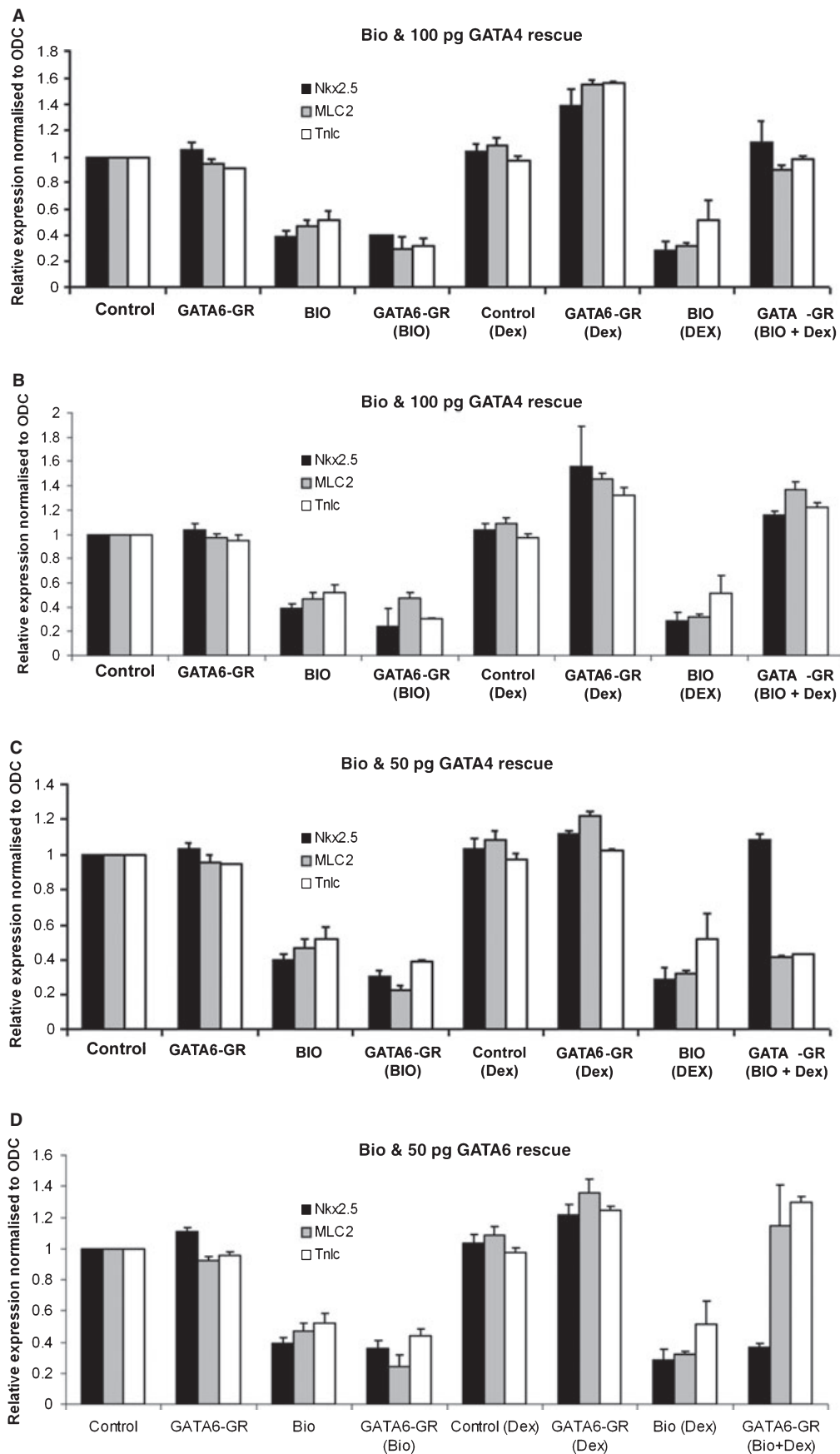


Fig. 6 The pharmacological Wnt/ β -catenin agonist BIO inhibits MLC2 expression at a concentration of 6 μ M. (A–H) Analysis of RNA *in-situ* hybridization of cardiac marker MLC2 in stage 32 embryos. Embryos were treated at stage 20 with BIO at 2, 4 and 6 μ M either alone (C,E,G) or in the presence of dexamethasone (D,F,H, respectively). (I) Bar charts showing the percentage of embryos with high, normal or low MLC2 expression. n, number of embryos assayed for each experimental and control manipulation; MMR, Mark's modified Ringers standard buffer.

transcription factors. We analysed our experiments using whole-embryo qPCR, which is more sensitive and which allowed us to analyse the gene expression levels of several additional markers. In control experiments we first confirmed that the analysis of Nkx2.5, MLC2 and Tnnc gene expression by qPCR (Fig. 5A,B) gave results consistent with analysis by RNA *in-situ* hybridization (see above). We then analysed these same validated RNA samples for gene expression of other pro-cardiogenic transcription factors, i.e. Nkx2.3 (Sparrow et al. 2000) and either GATA4 or

GATA6, as well as Wnt11b (Pandur et al. 2002; Garriock et al. 2007), which was recently identified as a direct target gene of GATA factors (Afouda et al. 2008). We found that activated β -cateninGR reduced the expression of Nkx2.3, *gata4*, *gata6* and Wnt11b (Fig. 5C,D). As expected, reinstating GATA4 (Fig. 5C) or GATA6 (Fig. 5D) function rescued the expression of Nkx2.3 and Wnt11b as well as *gata6* or *gata4*, respectively. Although Wnt11b was used here as a convenient GATA target control, the effects on cardiogenic Wnt11b expression are difficult to monitor with the qPCR

Fig. 7 Wnt/ β -catenin-mediated inhibition of cardiogenic gene expression is differentially rescued by GATA4 and GATA6 function. Analysis of gene expression for the cardiogenic transcription factors Nkx2.3, Nkx2.5, *gata4* or *gata6*, the GATA target gene Wnt11b and the cardiomyocyte differentiation markers MLC2 and Tnnc using qPCR in whole embryos at stage 32. Embryos were injected at the four-cell stage in the dorsal marginal zone of both blastomeres with either GATA4GR-encoding mRNA (A,E, 100 pg; C,G, 50 pg per blastomere) or GATA6GR-encoding mRNA (B,F, 100 pg; D,H, 50 pg per blastomere). Embryos were cultured from stage 20 where indicated with the Wnt agonist BIO (to activate the Wnt/ β -catenin pathway) and dexamethasone (to activate the GATAGR constructs). This experiment was carried out independently three times before all data were collated. The relative expression of the described cardiac markers was normalized to the housekeeping gene ODC and measured in comparison to control unmanipulated embryos normalized to equal 1 relative expression unit. Statistical analysis was then carried out using a parametric paired sample *t*-test.



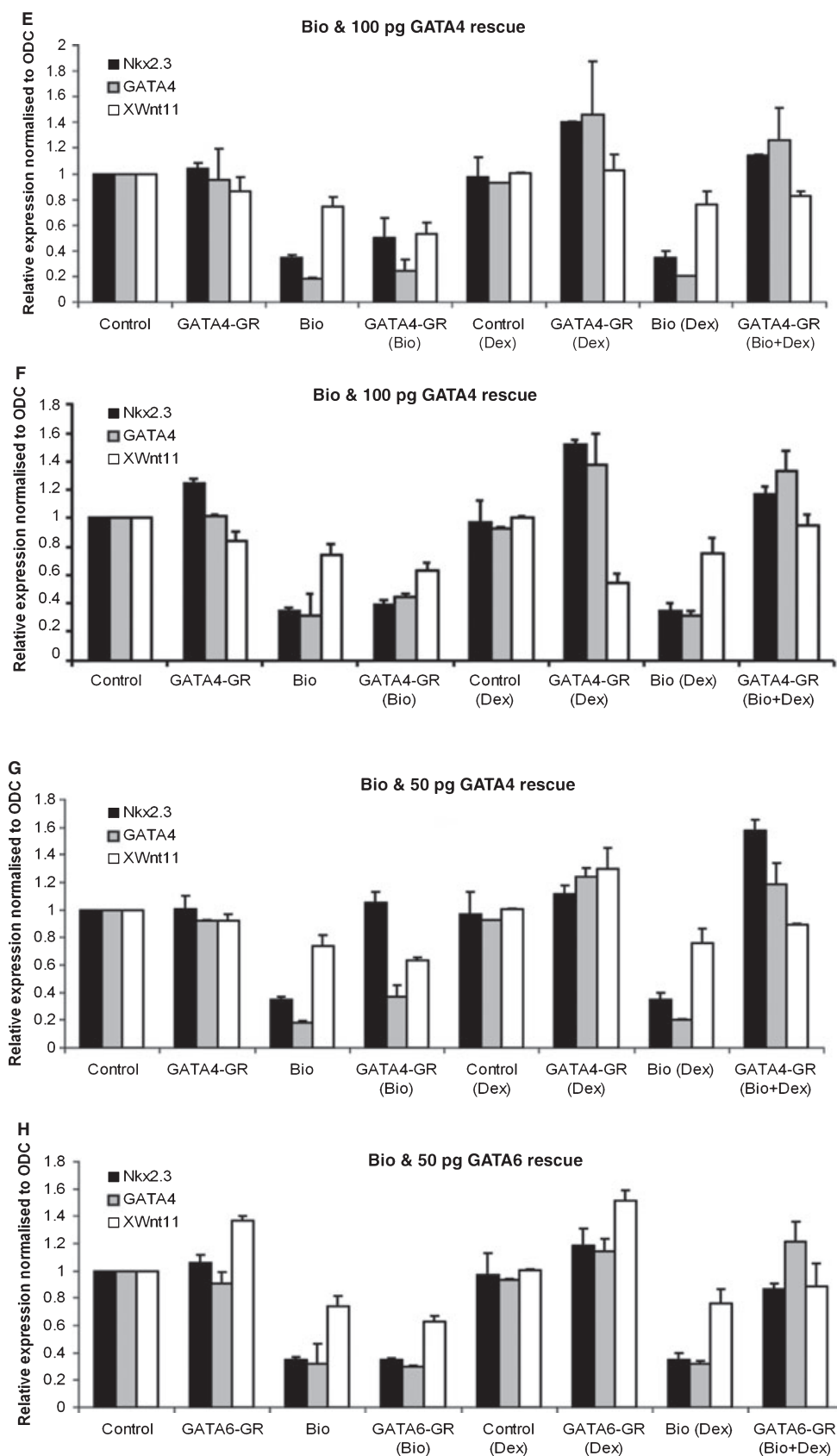


Fig. 7 (Continued)

method, as it will detect not only cardiogenic expression but also Wnt11b expression in other tissues where it is not under GATA regulation. However, this qPCR analysis of our rescue experiment confirms that reinstated GATA function does rescue expression of the pro-cardiogenic transcription factor network involving the well-studied Nkx2.5 and also the related factor Nkx2.3, the endogenous *gata4* and *gata6* genes, and the known GATA target Wnt11b.

However, careful analysis of this experiment suggests a subtle difference between the rescues with GATA4 and with GATA6. GATA4 appears to rescue the expression of the marker genes analysed in this experiment to a higher level than the rescue by GATA6 (compare Fig. 5C with 5D). Analysis with qPCR is expected to be more quantitative than the RNA *in-situ* hybridization analysis shown above, presumably allowing detection of this subtle difference between the GATA4- and GATA6-mediated rescues.

Functional differences between GATA4 and GATA6

We wanted to investigate in more detail the subtle functional difference between GATA4 and GATA6 suggested by qPCR analysis of the rescue experiments. Experiments using the agent BIO to cause general activation of Wnt/ β -catenin signalling resulted in remarkably similar effects on cardiomyogenesis as the more elaborate β -cateninGR-encoding mRNA injection experiments and therefore we used BIO in the following analysis. We first assayed the effect of different doses of BIO on subsequent heart muscle differentiation (Fig. 6) to determine the lowest dose of BIO that gave us clear inhibition of cardiomyogenesis. We also confirmed that the application of dexamethasone, to be used in subsequent experiments for the induction of GATA4GR or GATA6GR constructs, did not alter the effect of BIO. BIO at 6 μ M was ideal for activating Wnt/ β -catenin signalling sufficiently to inhibit cardiomyogenesis in the majority of embryos, irrespective of whether dexamethasone was added or not. In the following experiments we therefore combined inhibition of cardiogenesis (using the Wnt/ β -catenin signalling agonist BIO) with attempted rescue by dexamethasone-activated GATA4GR or GATA6GR (the mRNA of which had previously been injected at doses of 50 or 100 pg per blastomere).

We initially analysed the experiment with high doses of GATA4GR or GATA6GR mRNA by monitoring changes to marker gene expression with qPCR (Fig. 7A,B,E,F). The results confirmed our conclusions that the reinstatement of GATA function rescues cardiogenesis after inhibition by Wnt/ β -catenin signalling (see above). However, the experiments with lower doses of injected mRNA reveal differences between the extent of the rescue by GATA4GR and GATA6GR (Fig. 7C,D,G,H). These differences encouraged us to examine the gene expression of the pro-cardiogenic transcription factor Nkx2.5 and the structural proteins MLC2 and Tnnc using whole-mount RNA *in-situ* hybridiza-

tion (Figs 8 and 9). Both methods of analysis reveal that, under these experimental conditions, GATA4 rescues the expression of Nkx2.5 (Figs 7C and 8H,Y) but GATA6 does not (Figs 7D and 9H,Y). However, GATA6 is proficient at rescuing the expression of MLC2 and Tnnc (Figs 7D and 9P,X,Z,AA) but GATA4 is less able to do so (Figs 7C and 8P,X,Z,AA). These results show that GATA4 is more efficient at reinstating the expression of genes in the pro-cardiogenic transcription factor network (Nkx2.3 and Nkx2.5), whereas GATA6 is more potent at promoting the expression of genes associated with terminal differentiation (MLC2 and Tnnc).

This result reveals specific activities for the GATA4 and GATA6 constructs, arguing against simple quantitative differences between these constructs. It also seems likely that the subtle functional differences observed by qPCR (Fig. 5C,D) are therefore not due to technical efficiency differences between the GATA4GR and GATA6GR constructs. It may be useful to explore in future experiments whether other GATA gain-of-function experiments (e.g. using inducible transgenics) reveal similar activity differences between different GATA proteins. It may be that our experimental design was particularly prone to revealing these inherent activity differences because activated Wnt/ β -catenin signalling cleared endogenous *gata* expression completely before exogenous GATA proteins could be individually tested in rescue experiments for their specific functional activities during organogenesis stages. It may prove more difficult to discover functional differences between GATA proteins in a background where endogenous *gata* genes are still expressed.

The results of the GATA gain-of-function experiments reported here complement the conclusions of *gata* loss-of-function analyses of myocardium development (Peterkin et al. 2003, 2007; Haworth et al. 2008). It was shown that loss of *gata6* function did not affect the gene expression of Nkx2.3, Nkx2.5 or the related downstream pro-cardiogenic transcription factor Nkx2.10. Loss of *gata6* function did, however, affect the expression of heart muscle differentiation markers (MLC2 and cardiac actin; Peterkin et al. 2003), suggesting a primary role for *gata6* in differentiation (Peterkin et al. 2007). Such conclusions correspond well with our finding that reinstating GATA6 function particularly rescues markers associated with terminal differentiation but has less effect on the pro-cardiogenic transcription factors Nkx2.3 and Nkx2.5. Previous loss-of-function analyses also indicate that *gata4* functions in a mostly redundant role in a self-reinforcing evolutionarily conserved feedback loop during induction of cardiomyogenesis (Peterkin et al. 2007), notwithstanding an even earlier requirement for the related gene *gata5* for the initial development of heart precursors (Haworth et al. 2008). This different requirement for *gata4* may reflect the GATA4 activity discovered in our rescue experiments, which caused greater induction of Nkx2.3 and Nkx2.5 expression than of terminal differentiation markers.

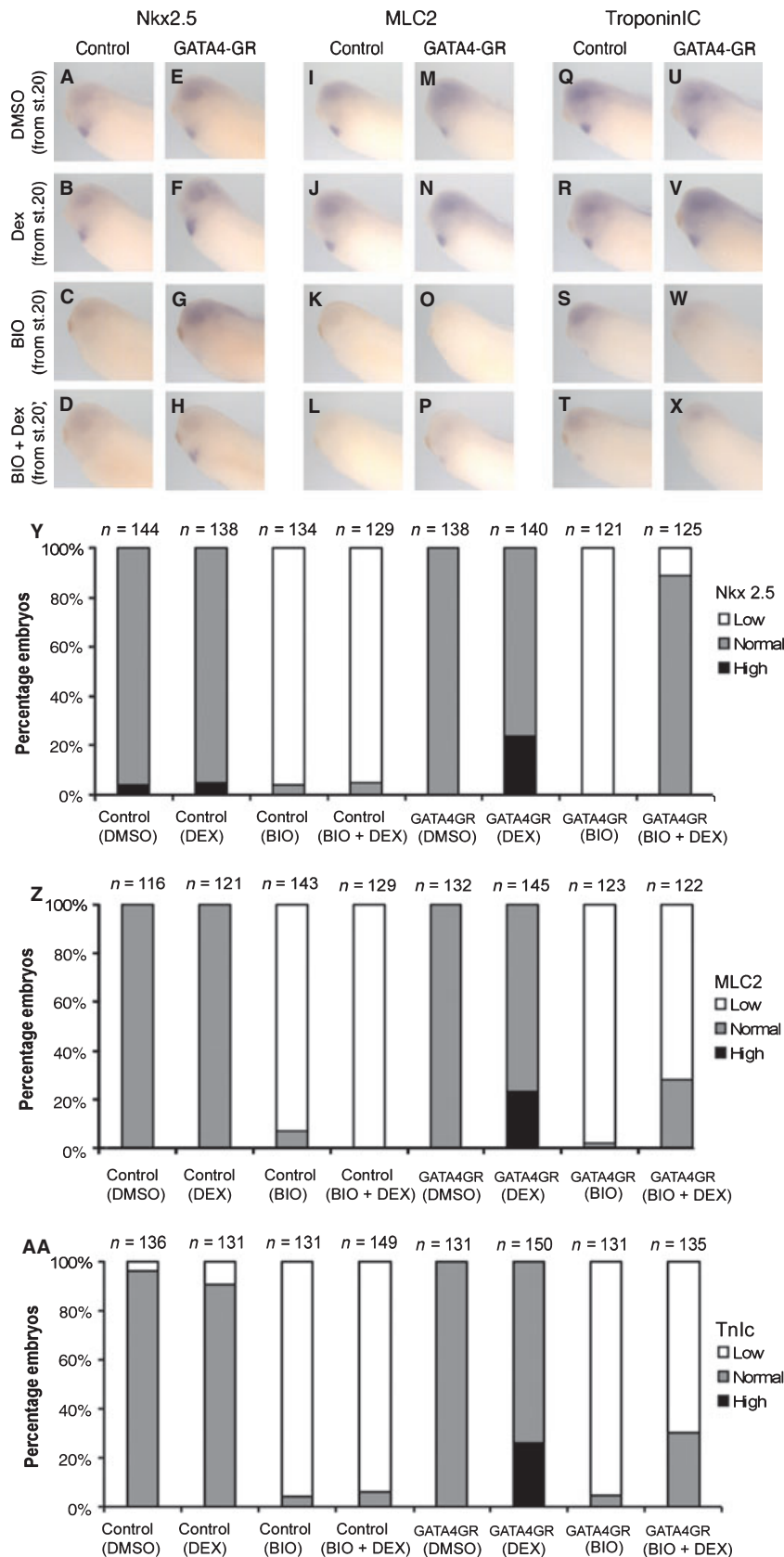


Fig. 8 GATA4 preferentially rescues the expression of early cardiogenic transcription factor genes. Analysis of Nkx2.5 (A–H), MLC2 (I–P) and Tnnc (Q–X) expression by whole-mount RNA *in-situ* hybridization at stage 32 in uninjected control (A–D, I–L, Q–T) and GATA4GR RNA-injected (50 pg per blastomere) (E–H, M–P, U–X) embryos, which were treated from stage 20 with dimethylsulphoxide (DMSO) as a control (A, E, I, M, Q, U), dexamethasone alone as a control (B, J, R) or to activate GATA4GR (F, N, V), BIO alone to activate the Wnt/ β -catenin pathway (C, G, K, O, S, W), and BIO and dexamethasone together (D, H, L, P, T, X) to activate Wnt/ β -catenin signalling and GATA4GR at the same time (H, P, X). Note the reduced gene expression of the analysed cardiogenic markers in embryos with BIO-mediated activated Wnt/ β -catenin signalling (G, O, W) and the restored expression of the cardiogenic transcription factor gene Nkx2.5 (H) when GATA4 function is restored. Bar charts showing the percentage of embryos with high, normal or low Nkx2.5 (Y), MLC2 (Z) and Tnnc (AA) expression in the experiments illustrated in A–X. n, number of embryos assayed for each experimental and control manipulation.

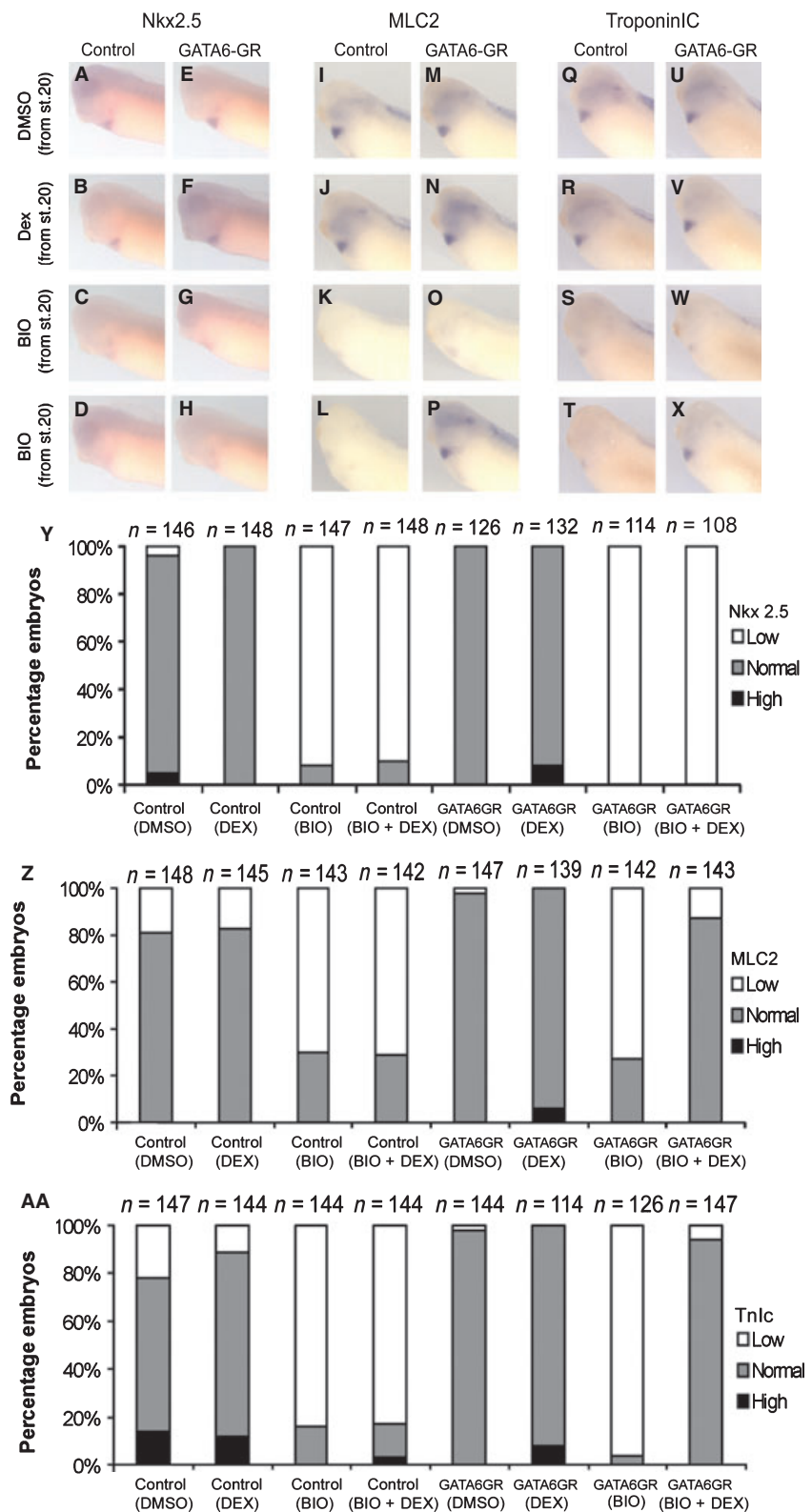


Fig. 9 GATA6 preferentially rescues the expression of cardiomyogenic differentiation genes. Analysis of Nkx2.5 (A–H), MLC2 (I–P) and Tnnc (Q–X) expression by whole-mount RNA *in-situ* hybridization at stage 32 in uninjected control (A–D, I–L, Q–T) and GATA6GR RNA-injected (50 pg per blastomere) (E–H, M–P, U–X) embryos, which were treated from stage 20 with dimethylsulphoxide (DMSO) as a control (A, E, I, M, Q, U), dexamethasone alone as a control (B, J, R) or to activate GATA6GR (F, N, V), BIO alone to activate the Wnt/ β -catenin pathway (C, G, K, O, S, W), and BIO and dexamethasone together (D, H, L, P, T, X) to activate Wnt/ β -catenin signalling and GATA6GR at the same time (H, P, X). Note the reduced gene expression of the analysed cardiogenic markers in embryos with BIO-mediated activated Wnt/ β -catenin signalling (G, O, W) and the restored expression of the cardiomyogenic differentiation genes MLC2 (P) and Tnnc (X) when GATA6 function is restored. Bar charts showing the percentage of embryos with high, normal or low Nkx2.5 (Y), MLC2 (Z) and Tnnc (AA) expression in the experiments illustrated in A–X. n, number of embryos assayed for each experimental and control manipulation.

Our results suggest GATA activity differences between initial cardiogenic induction and subsequent cardiomyocyte differentiation but our experiments were not designed to

directly address activity differences at different embryonic stages. Our experiments also do not rule out cell-non-autonomous or tissue-non-autonomous differences

between GATA4 and GATA6 function. Our results are consistent with the idea that different GATA factors have specific target genes. It is currently unclear how such molecular activity differences between the different members of the GATA4, GATA5 and GATA6 protein family would come about, as they all bind with high affinity to canonical GATA cis-regulatory DNA binding sites, although some subtle differences in their binding site preferences have previously been reported (Sakai et al. 1998). In addition, the *gata6* gene encodes GATA6 transcription factor proteins that have unique N-terminal protein domains unlike GATA4 or GATA5 (Brewer et al. 1999; Peterkin et al. 2003; Afouda et al. 2005). These unique protein domains could mediate functionally important interactions with other transcription factors to bring about the specific activities that we discovered in our rescue experiments.

Conclusions

We show in whole-embryo experiments that the inhibition of heart development by Wnt/ β -catenin signalling during organogenesis stages is mediated by the loss of expression of pro-cardiogenic transcription factors, as reinstating pro-cardiogenic GATA factors rescues subsequent heart development. However, these results also reveal that different GATA factors primarily rescue either gene expression of other pro-cardiogenic transcription factors or terminal cardiomyocyte differentiation.

Author contributions

S.H. and B.A.A. conceived the project and secured PhD scholarship funding, S.H. and J.M. designed the individual experiments with input from B.A.A., J.M. carried out the experiments and analysis, and J.M. and S.H. wrote the manuscript with comments from B.A.A.

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